

$$\vec{x} = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}$$

$$Ax = b$$

$$\vec{b} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

MATICI A ZÍŠKA'M DOSAZEM'M 0,1,2 DO POLYNOMU

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \end{pmatrix} \vec{x} = A^T \cdot (A \cdot A^T)^{-1} \cdot \vec{b}$$

$$A \cdot A^T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \\ 0 & 1 & 8 \\ 0 & 1 & 16 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 31 \\ 1 & 31 & 341 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 5 & 31 \\ 0 & 0 & 1 & 1 & 31 & 341 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ -1 & 1 & 0 & 0 & 4 & 30 \\ -1 & 0 & 1 & 0 & 30 & 340 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ -\frac{1}{4} & \frac{1}{4} & 0 & 0 & 1 & \frac{15}{2} \\ -1 & 0 & 1 & 0 & 30 & 340 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ -\frac{1}{4} & \frac{1}{4} & 0 & 0 & 1 & \frac{15}{2} \\ \frac{13}{2} & -\frac{15}{2} & 1 & 0 & 0 & 115 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ -\frac{1}{4} & \frac{1}{4} & 0 & 0 & 1 & \frac{15}{2} \\ \frac{13}{230} & -\frac{3}{46} & \frac{1}{115} & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} -\frac{217}{230} & \frac{3}{46} & -\frac{1}{115} & 1 & 1 & 0 \\ \frac{-62}{92} & \frac{68}{92} & -\frac{3}{46} & 0 & 1 & 0 \\ \frac{13}{230} & -\frac{3}{46} & \frac{1}{23} & 0 & 0 & 1 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{ccc|ccc} \frac{186}{115} & -\frac{31}{46} & \frac{13}{230} & 1 & 0 & 0 \\ -\frac{31}{46} & \frac{17}{23} & -\frac{3}{46} & 0 & 1 & 0 \\ \frac{13}{220} & -\frac{3}{46} & \frac{1}{115} & 0 & 0 & 1 \end{array} \right)$$

$$(A \cdot A^T)^{-1}$$

$$A^T \cdot (A \cdot A^T)^{-1} = \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & \frac{186}{115} & -\frac{31}{46} & \frac{13}{230} \\ 0 & 1 & 2 & -\frac{31}{46} & \frac{17}{23} & -\frac{3}{46} \\ 0 & 1 & 4 & \frac{13}{220} & -\frac{3}{46} & \frac{1}{115} \\ 0 & 1 & 8 & & & \\ 0 & 1 & 16 & & & \end{array} \right) =$$

$$\begin{array}{r}
 \begin{array}{r}
 1 \quad 0 \quad 0 \\
 \hline
 -129 \quad 14 \quad -11 \\
 \hline
 230 \quad 23 \quad 230
 \end{array} \\
 \begin{array}{r}
 -103 \quad 11 \quad -7 \\
 \hline
 230 \quad 23 \quad 230
 \end{array} \\
 \begin{array}{r}
 -59 \quad 5 \quad 1 \\
 \hline
 230 \quad 23 \quad 230
 \end{array} \\
 \begin{array}{r}
 53 \quad -7 \quad 17 \\
 \hline
 230 \quad 23 \quad 230
 \end{array}
 \end{array} = \beta$$

$$\beta \cdot \vec{b} = \begin{array}{r}
 1 \quad 0 \quad 0 \\
 \hline
 -129 \quad 14 \quad -11 \\
 \hline
 230 \quad 23 \quad 230 \\
 -103 \quad 11 \quad -7 \\
 \hline
 230 \quad 23 \quad 230 \\
 -59 \quad 5 \quad 1 \\
 \hline
 230 \quad 23 \quad 230 \\
 53 \quad -7 \quad 17 \\
 \hline
 230 \quad 23 \quad 230
 \end{array}$$

$$\begin{array}{r}
 1 \\
 2 \\
 -3
 \end{array} = \begin{array}{r}
 1 \\
 4 \\
 5 \\
 3 \\
 5 \\
 1 \\
 5 \\
 -3 \\
 5
 \end{array}$$

H/LE DANT' POLYNOM JE

$$P(x) = \frac{3}{5}x^4 + \frac{1}{5}x^3 + \frac{3}{5}x^2 + \frac{4}{5}x + 1$$